

NAME: Dongfang Zhao

POSITION TITLE & INSTITUTION: Assistant Professor, University of Nevada, Reno

A. PROFESSIONAL PREPARATION - (see [PAPPG Chapter II.C.2.f.\(i\)\(a\)](#))

INSTITUTION	LOCATION	MAJOR/AREA OF STUDY	DEGREE (if applicable)	YEAR (YYYY)
KU Leuven	Belgium	Statistics	Diploma	2005
Emory University	Atlanta, GA	Computer Science	MS	2008
Illinois Inst. of Tech.	Chicago, IL	Computer Science	PhD	2015
Univ. of Washington	Seattle, WA	Computer Science	PostDoc	2016

B. APPOINTMENTS - (see [PAPPG Chapter II.C.2.f.\(i\)\(b\)](#))

From - To	Position Title, Organization and Location
2017 - now	Assistant Professor, University of Nevada, Reno
2009 - 2011	Software Developer, Epic Systems, Madison, Wisconsin

C. PRODUCTS - (see [PAPPG Chapter II.C.2.f.\(i\)\(c\)](#)) Products Most Closely Related to the Proposed Project

The PI is the corresponding author (i.e., the last author) of the following publications. The first authors of these publications (Abdullah Al-Mamun, Xinying Wang) are Ph.D. students under the PI's supervision.

[1] Abdullah Al-Mamun, Haoting Shen, and Dongfang Zhao. DEAN: A Lightweight and Resource-efficient Blockchain Protocol for Reliable Edge Computing. IEEE International Parallel & Distributed Processing Symposium (IPDPS). 2022.

[2] Abdullah Al-Mamun, Feng Yan, and Dongfang Zhao. BAASH: Lightweight, Efficient, and Reliable Blockchain-As-A-Service for HPC Systems. International Conference for High Performance Computing, Networking, Storage, and Analysis (SC). 2021.

[3] Abdullah Al-Mamun, Feng Yan, and Dongfang Zhao. SciChain: Blockchain-enabled Lightweight and Efficient Data Provenance for Reproducible Scientific Computing. The 37th IEEE International Conference on Data Engineering (ICDE). 2021.

[4] Xinying Wang, Abdullah Al-Mamun, Feng Yan, and Dongfang Zhao. Toward Accurate and Efficient Emulation of Public Blockchains in the Cloud. International Conference on Cloud Computing (CLOUD). 2019.

[5] Abdullah Al-Mamun, Tonglin Li, Mohammad Sadoghi, and Dongfang Zhao. In-Memory Blockchain: Toward Efficient and Trustworthy Data Provenance for HPC Systems. IEEE International Conference on Big Data (BigData). 2018.

Other Significant Products, Whether or Not Related to the Proposed Project

- [1] Xinying Wang, Olamide Timothy Tawose, Feng Yan, and Dongfang Zhao. HDK: Toward High-Performance Deep-Learning-Based Kirchhoff Analysis. The 34th AAAI Conference on Artificial Intelligence (AAAI). 2020.
- [2] Heyang Qin, Syed Zawad, Yanqi Zhou, Lei Yang, Dongfang Zhao, and Feng Yan. Swift Machine Learning Model Serving Scheduling: A Region-Based Reinforcement Learning Approach. International Conference for High Performance Computing, Networking, Storage, and Analysis (SC). 2019.
- [3] Parmita Mehta, Sven Dorkenwald, Dongfang Zhao, Tomer Kaftan, Alvin Cheung, Magdalena Balazinska, Ariel Rokem, Andrew Connolly, Jacob Vanderplas, and Yusra AlSayyad. Comparative Evaluation of Big Data Systems on Scientific Image Analytics Workloads. The 43rd International Conference on Very Large Databases (VLDB). 10 (11):1226--1237, 2017.
- [4] Dongfang Zhao, Ning Liu, Dries Kimpe, Robert Ross, Xian-He Sun, and Ioan Raicu. Towards Exploring Data-Intensive Scientific Applications at Extreme Scales through Systems and Simulations. IEEE Transactions on Parallel and Distributed Systems (TPDS). 27(6):1824--1837, 2016.
- [5] Dongfang Zhao and Li Yang. Incremental Isometric Embedding of High-Dimensional Data Using Connected Neighborhood Graphs. IEEE Transactions on Pattern Analysis and Machine Intelligence (PAMI). 31(1):86--98, 2009.

D. SYNERGISTIC ACTIVITIES - (see [PAPPG Chapter II.C.2.f\(i\)\(d\)](#))

1. Editorial Services: The PI currently serves on the editorial board of the IEEE Transactions on Parallel and Distributed Systems, International Journal of Cooperative Information Systems, and the Springer Journal of Big Data.
2. Technical Review: The PI served on an NSF CISE grant proposal review panel in 2018 and 2022. The PI served on a DOE grant proposal review panel in 2018.
3. Organizing Conferences: The PI has been co-chairing the ACM/USENIX International Middleware Conference in Davis, California, 2019.
4. Software Contribution: In 2012-15, the PI led the design and implementation of an open-source distributed filesystem, namely FusionFS (published at TPDS'16). Since 2017, the PI has been an active contributor to various projects such as ImageBench (published at VLDB'17), BlockLite (published at CLOUD'19), HDK (published at AAAI'20), BASSH (published at SC'21), SciChain (published at ICDE'21), and Parma (published at IPDPS'22).
5. Curriculum Development: The PI is a founding faculty member of the Big Data Minor at the University of Nevada, Reno (UNR) in 2018. The PI developed a new graduate course CS 791: Distributed Computing, at UNR in 2020.